

P Yuvaraj

AWS Data Engineer | Generative AI Practitioner

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PROFESSIONAL SUMMARY

AWS Data Engineer with **3+ years** of experience building scalable **data pipelines, data lakes, and cloud-native solutions** using **AWS S3, Glue, Lambda, Athena, and CloudWatch**. Strong expertise in **ETL development, cloud migration, data quality, and performance optimization**. Building **Generative AI-augmented data engineering solutions** using **AWS Bedrock, Anthropic Claude, and Large Language Models (LLMs)** for intelligent data validation and automation. Proficient in **Python, SQL, PySpark**, and modern data engineering practices.

WORK EXPERIENCE

CAPGEMINI — AWS Data Engineer

11/2022 – Present | Bengaluru, India

Enterprise Data Integration (Informatica Intelligent Cloud Services - IICS)

- Developed **Extract, Transform, Load (ETL)** mappings and taskflows to extract, transform, and load data from multiple source systems into target databases.
- Monitored job executions, resolved failures, and implemented data validation rules to improve workflow performance and data quality.

Data Engineering & Pipeline Development

- Designed and developed scalable ETL pipelines using **AWS Glue, S3, and Lambda** to process structured and unstructured data.
- Built automated data workflows improving ingestion efficiency and reducing manual intervention.
- Implemented data validation and transformation logic ensuring high data accuracy across multiple environments.

Performance Optimization

- Optimized **Amazon Athena** queries to improve query performance and reduce execution costs.
- Enhanced AWS Glue job performance through partitioning and efficient data-handling techniques.
- Designed an optimized **S3-based data lake architecture** for better query performance and scalability.

Cloud Migration

- Led migration of on-premises datasets to AWS cloud using **S3, Glue, and Lambda**.
- Performed schema validation, reconciliation, and parallel-run comparisons ensuring seamless migration.

Automation & Generative AI Initiatives

- Engineered a **Python REST API automation framework** for QMetry/Jira test case and cycle creation — **~97% effort reduction, ~283 hours saved annually** across QA pods; formally approved for team-wide rollout.
- Developing an **AI-Powered Data Migration Validation Framework** using **AWS Bedrock and Anthropic Claude (LLM)** for schema inference and semantic mismatch interpretation, integrating with **Apache Airflow (MWAA)** pipelines.

PROJECT HIGHLIGHTS

AI-Powered Data Migration QA Validation Framework

AWS Bedrock, Anthropic Claude (LLM), Python, Athena, Glue, S3, Apache Airflow (MWAA)

- Architecting a config-driven validation framework automating source-to-target migration QA using **6 reconciliation checks**: DDL, record count, null, duplicate, date, and sample-data validation.
- Building a deterministic **SQL/pandas** engine for auditable pass/fail results, paired with a **Generative AI layer (AWS Bedrock + Anthropic Claude)** that infers test specifications from schemas and interprets format/semantic mismatches via **LLM-based judgment**, eliminating false-positive manual review.
- Applying core Generative AI concepts — **Prompt Engineering and LLM-augmented decision logic** — to transform manual QA into a scalable, self-service validation tool integrated with **Apache Airflow (MWAA)** pipelines.
- Improving data quality coverage and reducing manual validation effort through automated evidence capture, QA reporting, and alerting.

QMetric / Jira QA Workflow Automation Framework

Python, Jira REST API, QMetric API, Personal Access Token (PAT) authentication

- Engineered a Python automation framework for QMetric/Jira test case and test cycle creation, replacing a semi-manual Excel-driven workflow through **REST API integration**.
- Reduced test case creation time from **~15 minutes to ~30 seconds per table (~97% reduction)**, saving **~283 hours annually** across QA pods.
- Authored **Standard Operating Procedure (SOP)** and ROI analysis; formally approved by QA leadership for team-wide adoption.

TECHNICAL SKILLS

- **Cloud (AWS)**: Amazon S3, AWS Glue, AWS Lambda, AWS IAM, Amazon Athena, Amazon CloudWatch, Amazon MWAA (Managed Workflows for Apache Airflow), AWS Bedrock
- **Generative AI / AI Engineering**: AWS Bedrock, Anthropic Claude, Large Language Models (LLMs), Prompt Engineering, LLM-as-Judge patterns. *Familiar with*: Embeddings, Retrieval-Augmented Generation (RAG), Agentic Patterns
- **Data Engineering**: ETL/ELT Development, Data Migration, Data Lake Architecture, Data Quality, Informatica PowerCenter, IICS, Snowflake, PySpark, Apache Airflow
- **Programming**: Python, SQL, Unix/Linux Shell Scripting
- **Tools**: Git, Jira, QMetric, REST APIs

EDUCATION

Bachelor of Engineering (B.E.), Computer Science

Alva's Institute of Engineering and Technology — 2022